DIGIT-PARTITIONING PRIMES AND THE ALIGNMENT FORMULA

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Abstract. For a prime p not dividing a positional base b, the fractions $\{k/p : 1 \leq k \leq p-1\}$ partition into cosets of the subgroup $\langle b \rangle \leq (\mathbb{Z}/p\mathbb{Z})^*$, each coset determining a family of repetends related by cyclic permutation. We say that p is *digit-partitioning* in base b if no two fractions from the full set $\{k/p\}$ share a digit at any common repetend position unless they are identical.

We prove that p is digit-partitioning in base b if and only if $p \leq b+1$. The proof rests on the injectivity of the digit function $r \mapsto \lfloor br/p \rfloor$ on $\{1, \ldots, p-1\}$.

For every digit-partitioning prime, the repetend alignment of $n = pm$ (with m a b-smooth integer) equals exactly $(2m-1)/(pm-1)$. This universal formula, combined with the golden threshold analysis of the companion paper, confirms that the golden ratio selects $p = 3$ uniquely among primes $p \geq 3$: the alignment limit $2/p$ exceeds $1/\varphi$ only for $p = 2$ and $p = 3$.

1. Introduction

This is the second paper in a series studying the digit function $\delta(r) = \lfloor br/p \rfloor$ and its structural invariants.

In a companion paper [1], we proved that the repetend alignment of $n = 3m$ (with m a 10-smooth integer) equals $(2m-1)/(3m-1)$, and that the golden ratio is the unique self-referential threshold classifying such integers. The formula arose from the observation that non-matching fractions contribute zero position-wise alignment with the base repetend.

This paper identifies the exact condition under which zero cross-alignment holds for a general prime p in a general base b. The condition is $p \leq b+1$. When this holds, the alignment formula $(2m-1)/(pm-1)$ generalizes without modification. Among primes $p \geq 3$, the alignment limit $2/p$ exceeds $1/\varphi$ only for $p = 3$.

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2. Setting

Let $b \geq 2$ be a positional base and p a prime not dividing b. The multiplicative order $L = \mathrm{ord}_p(b)$ is the length of the repetend of $1/p$ in base b [4]. The group $(\mathbb{Z}/p\mathbb{Z})^*$ has order $p-1$, and the cyclic subgroup $\langle b \rangle$ has order L, partitioning $\{1, \ldots, p-1\}$ into $(p-1)/L$ cosets. Within each coset, the repetends of k/p are cyclic permutations of a common string.

Definition 1. An integer m is *b-smooth* if every prime factor of m divides b.

Definition 2. For $n = pm$ with m a b-smooth integer, the *repetend alignment* $\alpha(n)$ is the mean, over $k \in \{1, \ldots, n-1\}$, of the position-wise digit-match proportion between the repetend of k/n and the repetend of $1/n$. Terminating fractions contribute alignment 1.

3. The Digit Function

The central object is the digit function that maps remainders to digits during long division. The floor function $\lfloor br/p \rfloor$ partitions the integers $\{1, \ldots, p-1\}$ into contiguous bins of nearly equal size [2].

Definition 3. For a prime p not dividing b, the *digit function* is

$$\delta(r) = \left\lfloor \frac{br}{p} \right\rfloor, \qquad r \in \{0, 1, \ldots, p-1\}.$$

This is the digit produced when the remainder r is processed in one step of long division of any fraction with denominator p in base b.

Lemma 4. *If $p \leq b+1$, then δ is injective on $\{1, \ldots, p-1\}$.*

Proof. We treat two cases.

Case $p = b+1$. For $r \in \{1, \ldots, b\}$,

$$\delta(r) = \left\lfloor \frac{br}{b+1} \right\rfloor = \left\lfloor r - \frac{r}{b+1} \right\rfloor = r - 1,$$

since $0 < r/(b+1) < 1$ for $1 \leq r \leq b$. The map $r \mapsto r-1$ is a bijection from $\{1, \ldots, b\}$ to $\{0, \ldots, b-1\}$.

Case $p \leq b$. For distinct $r, s \in \{1, \ldots, p-1\}$ with $r < s$, the values br/p and bs/p differ by $b(s-r)/p \geq b/p \geq b/b = 1$. If $b(s-r)/p > 1$, the floors are necessarily distinct. If $b(s-r)/p = 1$ (which requires $s - r = p/b$, possible only when $b \mid p$, contradicting $p \nmid b$ for prime $p \leq b$), the values straddle a unit boundary unless both are integers, but br/p is never an integer for $1 \leq r < p$ since $\gcd(b, p) = 1$. In all cases, $\delta(r) \neq \delta(s)$. $\square$

Remark 5. When $p > b+1$, the digit function maps $p-1 > b$ distinct remainders into b digit values. By the pigeonhole principle [3], δ cannot be injective, and collisions are unavoidable.

4. The Digit-Partitioning Property

Definition 6. A prime p is *digit-partitioning* in base b if no two distinct fractions in $\{k/p : 1 \le k \le p-1\}$ share a digit at any common repetend position, unless their repetends are identical.

Theorem 7. *Let p be a prime not dividing b. Then p is digit-partitioning in base b if and only if $p \le b+1$.*

Proof. At position j of the repetend of k/p, the digit is $\delta(b^j k \bmod p)$. For distinct fractions k/p and ℓ/p with $k \not\equiv \ell \pmod p$, the arguments $b^j k$ and $b^j \ell$ are distinct modulo p (since b^j is invertible).

If $p \le b+1$, then δ is injective on $\{1, \ldots, p-1\}$ by Lemma 4, so distinct arguments produce distinct digits at every position. Therefore p is digit-partitioning.

If $p > b+1$, then δ maps $p-1 > b$ remainders into b values. By the pigeonhole principle, there exist distinct $r, s \in \{1, \ldots, p-1\}$ with $\delta(r) = \delta(s)$. The first repetend digit of r/p equals the first repetend digit of s/p, so p is not digit-partitioning. □

5. The Universal Alignment Formula

Proposition 8. *For $n = pm$ with m a b-smooth integer and p not dividing b, the non-terminating fractions in $\{k/n : 1 \le k \le n-1\}$ produce exactly $p-1$ distinct repetends.*

Proof. The $m-1$ multiples of p in $\{1, \ldots, pm-1\}$ give terminating fractions. The remaining $(p-1)m$ values of k are not divisible by p.

Choose t with $m \mid b^t$, and write $b^t = mu$ where $\gcd(u, p) = 1$ (possible since m is b-smooth and $p \nmid b$). Then

$$\frac{k}{pm} = \frac{ku}{p \cdot b^t},$$

so after t base-b digits, the repetend of $k/(pm)$ is exactly the repetend of $(ku \bmod p)/p$. Since multiplication by u permutes $(\mathbb{Z}/p\mathbb{Z})^*$, the non-terminating repetends of $\{k/(pm)\}$ are in bijection with the repetends of $\{r/p : 1 \le r \le p-1\}$.

Two distinct purely periodic fractions r/p and s/p with $r \not\equiv s \pmod p$ cannot have the same repetend (their digit sequences differ at position 0 when $\delta(r) \ne \delta(s)$, or at a later position by the invertibility of b modulo p). There are therefore exactly $p-1$ distinct repetends.

Each residue class modulo p contains exactly m elements of $\{1, \ldots, pm-1\} \setminus p\mathbb{Z}$, so each repetend is shared by exactly m fractions. □

Theorem 9. *If $p \leq b+1$ and $m \geq 2$ is b-smooth, then*

$$\alpha(pm) = \frac{2m-1}{pm-1}.$$

Proof. By Lemma 8, the $pm-1$ fractions partition into:

(A) $m-1$ terminating fractions (multiples of p). Each contributes alignment 1.
(B) m fractions whose cycle matches $1/(pm)$ (one of the $p-1$ cycle classes). Each contributes alignment 1.
(C) $(p-2)m$ fractions in the remaining $p-2$ cycle classes. Since p is digit-partitioning, distinct cycles share no digit at any common position (Theorem 7). Each contributes alignment 0.

Total: $(m-1) + m = 2m-1$ aligned out of $pm-1$. □

Corollary 10. *For fixed p, the alignment $\alpha(pm) \to 2/p$ as $m \to \infty$.*

6. Enumeration

For each base b, the digit-partitioning primes are precisely the primes $p \leq b+1$ that do not divide b.

Base b	Digit-part. primes ($p \leq b{+}1$, $p \nmid b$)	Alignment limit $2/p$
4	$3, 5$	0.667, 0.400
6	$5, 7$	0.400, 0.286
8	$3, 5, 7$	0.667, 0.400, 0.286
10	$3, 7, 11$	0.667, 0.286, 0.182
12	$5, 7, 11, 13$	0.400, 0.286, 0.182, 0.154
16	$3, 5, 7, 11, 13, 17$	0.667, ..., 0.118

Table 1. Digit-partitioning primes for selected bases and their alignment limits $2/p$.

In base 10, the digit-partitioning primes are 3, 7, and 11. These correspond to the three classical mechanisms identified in earlier analysis:

- $p = 3$: $\mathrm{ord}_3(10) = 1$ (single-digit repetends).
- $p = 7$: $\mathrm{ord}_7(10) = 6 = p-1$ (cyclic number, primitive root).
- $p = 11$: $\mathrm{ord}_{11}(10) = 2$, with $10 \equiv -1 \pmod{11}$ (nines-complement partition).

These mechanisms are unified by the single condition $p \leq b+1$.

7. Connection to the Golden Ratio

The alignment limit $2/p$ exceeds $1/\varphi$ only when $p < 2\varphi \approx 3.236$. Among primes $p \geq 3$, only $p = 3$ meets this condition. (For $p = 2$, the alignment limit is 1, trivially above any threshold.)

The self-referential characterization of [1] applies: the golden ratio's minimal polynomial $\tau^2 + \tau - 1$ divides the self-referential cubic if and only if $p = 3$, producing the fourth-power identity $\tau^4 = 2 - 3\tau$.

The present paper strengthens this result. The formula $(2m-1)/(pm-1)$ is not a special case tied to base 10 or to any particular mechanism. It holds universally for all digit-partitioning primes in all bases, and the characterization $p \leq b + 1$ shows that the class of such primes is determined by a single inequality. Among primes $p \geq 3$ in this class, $p = 3$ is the unique value with alignment limit above $1/\varphi$.

8. Remarks

Structural manifestations. Although the characterization is a single inequality, the digit-partitioning property manifests through different structural mechanisms depending on the relationship between b and p:

- When $b \equiv 1 \pmod{p}$: repetends have length one, and each fraction uses a unique single digit.
- When $p = b+1$ (so $b \equiv -1 \pmod{p}$): repetends have length two, and cosets correspond to $(b-1)$-complement pairs, partitioning the digit set completely.
- When b is a primitive root modulo p and $p \leq b+1$: there is a single coset, and the repetend is a cyclic number with all distinct digits.

These are consequences of the injectivity of δ, not independent conditions.

The boundary case $p = b + 1$. The case $p = b + 1$ is the tightest: δ maps b remainders into b digit values, and injectivity holds by a narrow margin. In base 10, this gives $p = 11$, whose five nines-complement pairs partition all ten digits exactly. The boundary case always produces the complement-pair structure, since $b + 1 = p$ implies $b \equiv -1 \pmod{p}$.

Open questions. For $p > b + 1$, the alignment formula acquires correction terms from digit collisions. Computing these corrections in closed form would extend the theory to all primes. That is the work of the papers that follow.

References

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